Research Highlights

- **First controlled ML–LLM comparison** for programming verdict classification using only submission metadata (no source code required), evaluated under identical zero-shot protocols across Random Forest, Gradient Boosting, Logistic Regression, DeepSeek-V3, and Qwen2.5:3B. Full zero-shot prompt template disclosed in Reproducibility Statement.

- **Platform-encoding boundary**: Compile-time errors (CE), time limit exceeded (TLE), and memory limit exceeded (MLE) (18.4% of submissions) are deterministically recoverable from execution metadata (F1 $\geq$ 0.89) because platform measurement thresholds define these outcomes. The WA$\leftrightarrow$RE boundary is not metadata-separable (RE F1 = 0.52), indicating where source-code analysis may be required.

- **Supervised ML outperforms frontier LLMs**: Gradient Boosting (95.02%) exceeds DeepSeek-V3 zero-shot (76.65%) by 18.4 pp, and Qwen2.5:3B (35.50%) by 59.5 pp. Even with identical features, GB (92.0%) surpasses DeepSeek-V3 by 15.4 pp. Execution time is the dominant predictor (8.8 pp accuracy drop when removed).

- **Statistical and empirical rigor**: Holm-Bonferroni-corrected McNemar tests, feature ablation, Gini importance, leave-one-user-out cross-validation (82.5% accuracy), cross-difficulty generalization (71–94% across problem difficulties), and user-disjoint replication confirming robustness to user-level clustering and look-ahead leakage (GroupShuffleSplit; GB 92.2%, RF 90.9% with zero train/test user overlap).

- **Open dataset, code, and deployment guidelines**: 13,360 Codeforces submissions with full metadata; deployment guidelines balancing accuracy, cost ($0.01/1K), latency (2 ms), and explainability across six educational contexts. Available at: `https://github.com/huwenbin123huwenbin/programming-error-classification`